

Drawing Lines to Connect

A Multi-Participant XR Audiovisual Art Experience Using Internet-Connected Video See-Through MR
with EEG/PPG

Ryu Nakagawa
Nagoya City University
Nagoya, Japan
nakagawaryu@sda.nagoya-cu.ac.jp

Hidefumi Ohmura
Tokyo University of Science
Tokyo, Japan
ohmura@is.noda.tus.ac.jp

Kenta Hidaka
Nagoya City University
Nagoya, Japan
c255719@ed.nagoya-cu.ac.jp

Sho Kato
Nagoya City University
Nagoya, Japan
c245709@ed.nagoya-cu.ac.jp

Ai Kawaguchi
Nagoya City University
Nagoya, Japan
c225008@ed.nagoya-cu.ac.jp

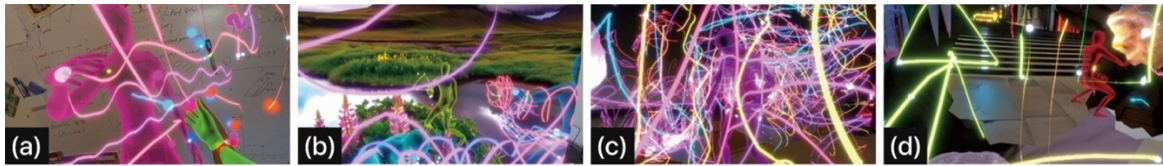


Figure 1: (a) A participant composing drawn lines in Edit mode. (b) A VR-Scape—an immersive environment based on AI-generated 360-degree imagery—emerges from the collision of two moving point objects following the drawn lines. (c) A participant situated within a dense array of lines drawn by four users, interwoven across real space and multiple VR-Scapes of varying scale. (d) Overlapping VR-Scapes. As multiple scenery windows remain in the physical environment, participants may gradually lose the sense of which layer corresponds to the “real” space, leading to a perceptual illusion of space.

Abstract

Participants wear a Meta Quest 3 headset and a Muse 2 device to freely draw lines in mid-air with their fingertips (see Figure 1). When moving points generated along these lines collide, piano sounds are generated with pitches corresponding to their velocities. At the same time, when certain conditions are met, AI-generated 360-degree virtual environments (VR-Scapes) appear and are shared among all participants (see Figures 1). The colors of avatars and drawn lines continuously shift according to an Emotion Color Map derived from physiological signals. Through this process, participants leave emotional and bodily traces within the shared MR space. As these traces overlap and intertwine, a multilayered form of “connection” emerges—among participants, between lines, and between real and virtual spaces—generating an evolving audiovisual artwork. While inspired by psychophysiological models, this project does not aim for quantitative emotion measurement at its current stage, but rather pursues an aesthetic exploration of emotion, embodiment, and connectivity within a shared mixed-reality experience.

CCS Concepts

• Applied computing; • Arts and humanities; • Media Arts;

ACM Reference Format:

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1 Introduction

Since 2017, we have been developing multi-participant interactive audiovisual artworks using mixed reality (MR) devices, based on the concept of “drawing sound in real space.” This project builds upon one of our representative works, *Drawing Sound in MR Space (DSMR)* (see Figure 2) [Nakagawa et al., 2018]. In DSMR, participants use an optical see-through MR device, the HoloLens, to draw lines in mid-air. Once a line is drawn, a moving point object is generated, which repeatedly travels from the starting point to the endpoint along the drawn trajectory. When two such objects collide, audiovisual effects—composed of musical fragments and particle-based visual elements—are triggered.

Through the collaborative drawing and editing of lines, participants engage in acts of creativity that give rise to evolving relationships within the shared MR space.

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Figure 2: (a) A participant experiencing DSMR. (b) Scene from a DSMR experience in progress.

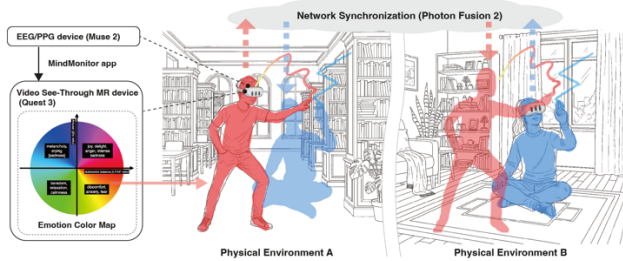


Figure 3: Technical implementation of “Drawing Lines to Connect.”

2 Motivation

Building on DSMR, this project introduces emotion estimation, generative AI, and a networked experience using a video see-through MR device, in order to pursue new artistic explorations in XR.

- In DSMR, the lines drawn in mid-air can be understood as bodily “traces.” By linking these traces with physiological signals such as EEG and PPG, we aim to extend them into expressions that reflect the emotional and temporal uniqueness of the present moment.
- During the COVID-19 pandemic, VR metaverse platforms provided new modes of communication. However, not all users felt comfortable within these fully virtual environments. Instead, the significance of one’s own physical surroundings was re-evaluated. In this context, reconstructing DSMR as an Internet-based MR experience using networked VST-MR offers a means of exploring new forms of co-creative communication that are grounded in each participant’s real-world environment.

3 Overview of The Work

Participants wearing a Meta Quest 3 headset and a Muse 2 device draw lines in the air (see Figure 3). Each completed line spawns a moving point object that travels repeatedly between its endpoints. When two moving points collide, a piano sound is triggered—the pitch determined by the combined velocities. Simultaneously, when certain conditions are met, AI-generated 360° VR-Scapes appear and disappear, synchronously shared across all participants. Avatars and lines change color in real time according to the participant’s current physiological state, linking emotional and bodily traces across networked physical spaces.

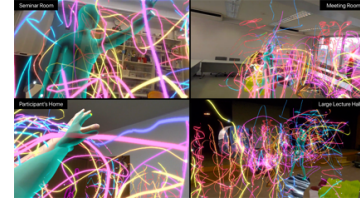


Figure 4: Demonstration of an Internet-Connected MR Experience Across Four Asymmetrical Physical Environments.

4 Implementation

4.1 Emotion Color Map

Following prior studies [Pope et al., 1995; Shaffer & Ginsberg, 2017; Kato & Murashita, 2022], EEG β/α ratio and PPG LF/HF ratio are mapped onto a two-dimensional affective space (see Figure 3), where the horizontal axis represents autonomic balance (LF/HF from PPG) and the vertical axis represents arousal (β/α from EEG). The resulting states are visualized as hue changes in the avatars and the drawn lines. This mapping is not intended as a scientifically validated measurement but rather as an artistic framework for visualizing affective dynamics. Hue values derived from this map continuously modify the color of lines and avatars, embodying emotion as a dynamic visual continuum.

4.2 AI-Generated VR-Scapes

A set of 360-degree images was pre-generated using Skybox AI with prompts designed via ChatGPT. Each image corresponds to an emotional quadrant in the Emotion Color Map. When moving points collide, the system selects a VR-Scape based on the average hue at the collision point and displays it for all participants. Here, the term “AI-generated” refers to pre-generated materials created through AI tools, not to real-time scene generation.

Previous work by Hachisu et al. [Hachisu et al., 2023] proposed a system that generates 360-degree environments from real-time conversations using generative AI. While this approach would also be ideal for our project, we aimed to display the VR-Scapes instantly; therefore, real-time generative AI processing was not implemented in the current system.

5 Outlook

The current version employs a single piano scale, pre-generated visuals, and simplified avatars. Nevertheless, demonstrations successfully connected four asymmetrical physical environments through the internet-connected MR (see Figure 4). As an art-focused exploration, no formal user study has yet been conducted; the value lies in co-creative engagement and aesthetic experience. Future work will include collaboration with neuroscientists to evaluate the effectiveness of emotion visualization within shared MR contexts, and to explore new forms of communication emerging within evolving audiovisual spaces.

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